

Impacts of soil compaction and historical soybean variety growth on soil macropore structure

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ABSTRACT

Soil compaction is the main form of soil degradation in agricultural systems. It leads to decreased infiltration and limited root growth. Plants can alleviate soil compaction through various mechanisms connected to root growth. However, it remains unclear whether plant breeding can improve the root traits of modern cultivars and their ability to alleviate compacted soil. The aim of this study was to determine the responses of soybean (*Glycine max* L. Merr.) roots to soil compaction and to observe the difference in soil alleviation by using soybean varieties bred before and after the development of mechanical agriculture in the Mollisol region of Northeast China. Five soybean varieties, bred from 1941 to 2003, were planted and subjected to either soil compaction or no compaction. Root traits and soil macropores were characterised by screening the root system (Shovelomics method) and by adopting dye-tracing methods. The properties of roots in soybean varieties showed noteworthy differences, although no clear breeding trends were detected. Compared with no compaction, the root lengths of all varieties decreased significantly when grown in compacted soil. Moreover, soil compaction decreased the vertical macroporosity and increased the heterogeneity of horizontal macropores. The modern soybean variety alleviated soil compaction by enabling the creation of more macropores, which were located deeper in the soil, compared to the older varieties. Overall, root growth was responsible for approximately one-third of macropore changes observed under compacted soil conditions, and the alleviation effects were greater in roots growing vertically than those observed in roots growing laterally. The observed variability of soil compaction alleviation among different soybean varieties could be considered in agriculture by breeders to select the varieties which decrease soil compaction.

1. Introduction

Soil compaction is an increasingly common phenomenon, associated with the intensive use of agricultural equipment and is recognised as one of the main threats to sustainable agriculture in the 21st century (Keller et al., 2019; Schjønning et al., 2015). Mechanical soil compaction leads to the loss of soil macropores and pore connectivity, and this is attributable to vertical compression and lateral shearing (Horn and Fleige, 2009; Schjønning et al., 2015). Thus, soil compaction may reduce water infiltration, increase the penetration resistance of plant roots, negatively impact the uptake of water and nutrients by plants, and finally, restrict plant growth (Bengough et al., 2011; Whalley et al., 2005; Whitmore

and Whalley, 2009).

The resistance of roots to soil compaction differs among plant species and varieties. Normally, tap-rooted crops are more capable of penetrating compacted soil than the fibrous-rooted ones (Chen and Weil, 2010; Pulido-Moncada et al., 2021; Ren et al., 2019b). Root growth ability in compacted soil even varies among crop varieties. For instance, differences in the penetrative abilities of roots have been reported in varieties of rice (Clark, 2008; Clark et al., 2002; Price et al., 2000; Samson et al., 2002; Zheng et al., 2000), wheat (Acuña et al., 2012; Botwright Acua et al., 2007; Kubo et al., 2006; Kubo et al., 2004; Kubo et al., 2008) and maize (Bushamuka and Zobel, 1998; Chimungu et al., 2015). Root penetration ability could vary ~ 10 times among rice

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cultivars (Clark et al., 2002), seven times in wheat (Kubo et al., 2006) and more than 3 times in maize (Chimungu et al., 2015). Overall, the response of roots to soil compaction among distinct species and varieties of plants remains poorly understood. For soybean varieties, there are even fewer reports.

Compared with root resistance to soil compaction, there are fewer studies focussed on the recovery of compacted soil by root growth. Roots can improve the quality of compacted soil through the rearrangement of soil particles during the penetration and expansion of roots or through creation of cracks mediated by water uptake (Ren et al., 2019b; Whalley and Bengough, 2013). Studies have revealed considerable variation in the ability of roots to alleviate soil compaction among species. For instance, Chen et al. (2014) found that tap-rooted Brassica cover crops possessed a substantial ability to alleviate compacted soil by increasing the least limiting water range and soil air permeability. Similarly, Pulido-Moncada et al. (2020, 2021) compared the ability of recovered compacted soil among different cover crop species, and the results showed that chicory (*Cichorium intybus*) and lucerne (*Medicago sativa* L.) significantly improved compacted soil structure by increasing macroporosity and connected small pores, compared with spring barley (*Hordeum vulgare* L.). However, to the best of our knowledge, no studies have compared the alleviation ability of soybean varieties grown in compacted soil.

Plant breeders have markedly improved soybean genetic traits responsible for adaptation to biotic and abiotic stress, and increased grain yield in the past decades; however, the genes responsible for these traits have seldom been considered to improve root traits because of the involvement of increased expenditure and challenging screening methods (Fried et al., 2018; Li et al., 2019). Interestingly, root traits are often unintentionally selected during historical improvements. For instance, modern soybean varieties showed larger root density (Boyer et al., 1980) and longer roots (Li et al., 2019) than the old varieties. Additionally, breeding times may also affect the alleviation effects because of the changing soil conditions (e.g., soil compaction levels) (Correa et al., 2019). However, these effects are rarely disentangled, especially among historical soybean varieties.

This study aimed to evaluate the resistance of historical soybean varieties to compacted soil and their resilience to changes in the soil macropore structure. The specific objectives were to 1) evaluate the differences in root traits and structure among different breeding time soybean varieties, 2) identify the main root traits that were sensitive to soil compaction, and 3) determine the relative impact of soil compaction and root alleviation on soil structure. We have hypothesised that soil compaction and root traits interactively affect the formation of soil structure. Moreover, breeding may enhance the soil alleviation effect of modern soybean varieties by improving root traits. To examine these hypotheses, the root traits of five soybean varieties, bred from 1941 to 2003, were compared and the soil macropore structure was determined under compaction and no compaction conditions.

2. Materials and methods

2.1. Study site

The study field locates at the experimental station of the Northeast Agricultural University (E 126.92 °, N 45.77 °), Harbin, Heilongjiang Province, China. The study area has a typical temperate continental monsoon climate, with an annual mean air temperature of 1.5 °C, a mean annual rainfall of 550 mm, and a frost-free period of approximately 120 days. The air temperature and precipitation distribution during the growing season are depicted in Supplementary Fig. 1. The studied soil was developed from loess-like parent materials and classified as Mollisol according to the USDA Soil Taxonomy (Soil Survey Staff, 2014). Soil samples were collected randomly to measure soil texture using the sieve-pipette method (Gee and Bauder, 1986) and to assess soil chemical properties (Table 1), using routing methods reported by Jiang et al. (2018).

2.2. Experimental design

A randomised split block design with three replicates was considered to determine the effects of soil compaction and soybean varieties in a relatively flat field. The soil was ploughed to a depth of 25 cm using a mouldboard, and then crushed to a depth of 15 cm with a rotary, before the conduction of soil compaction treatments. Soil compaction and no compaction treatments were laid out as main blocks and 25 soybean varieties were placed as subplots. Each block was 3 m wide × 22.5 m long, and each subplot was 3 m wide × 0.9 m long (Supplementary Fig. 2). Soil compaction was performed at approximately 60% of the field capacity, after six passes of John Deer 904 tractor, which had a curb weight of 4.6 Mg and resulted in a maximum ground pressure of 170 kPa (John Deer Tiantuo Co., Ltd. Tianjin, China). This procedure was performed on 12th May 2017. One day after the compaction, the compacted soil was tilled to a depth of 15 cm using a rotary to prepare for sowing soybeans. Seeds were sown (15 cm plant spacing, 30 cm interrow spacing, and 6 cm sowing depth) in each plot, and mixed mineral fertilisers (N:P:K of 13:5:10) were applied at a rate of 300 kg ha⁻¹ along lines 5 cm away from the plant rows.

Thereafter, the soil penetration resistance was measured at the depth of 60 cm using a hand-held penetrometer (Eijkelkamp Soil & Water). The penetrometer was equipped with a cone which had a base area of 1 cm² and a top angle of 60 °. Ten measurements were recorded randomly in each plot. Two plots that were not cropped were dug in the compaction and no compaction treatment surfaces to collect soil cores (100 cm³). This was performed to measure soil bulk density, with four replicates every 10 cm depth until to 60 cm depth in each profile.

2.3. Root traits

Soybean root traits were measured following the protocol of Shovelomics, which was first proposed by Trachsel et al. (2011) and later developed by Das et al. (2015). The method is based on the mapping of

Table 1
Selected soil properties before soil compaction treatment.

Depth (m)	Soil particle distribution (g kg ⁻¹)				pH (H ₂ O)	SOM ^b (g kg ⁻¹)	Total N (g kg ⁻¹)	Olsen P (mg kg ⁻¹)	Available K (mg kg ⁻¹)
	> 20 μm	20-2 μm	< 2 μm	Texture ^a					
0.0-0.1	58.40	17.47	23.33	Sandy clay loam	6.02	31.7	1.23	32.94	128.09
0.1-0.2	59.66	16.81	24.65	Sandy clay loam	6.19	30.7	1.21	39.67	123.39
0.2-0.3	57.50	19.10	24.71	Sandy clay loam	6.22	26.6	1.08	26.28	121.29
0.3-0.4					6.61	17.8	0.75	18.02	101.89
0.4-0.5					6.79	16.1	0.68	21.49	97.69
0.5-0.6	56.31	19.84	25.34	Sandy clay	6.78	15.4	0.60	30.81	102.49

^a According to the World Reference Base classification, FAO.

^b SOM, soil organic matter.

root morphology of legumes using an automated computing platform, i. e., Digital Imaging of Root Traits (DIRT, <http://dirt.cyverse.org/>). Five plants from each subplot were randomly collected at the full flowering stage and were considered as replicates for each variety. The above-ground biomass of all plants was removed from the cotyledon node using a pair of scissors. The underground volume was excavated carefully using a shovel, and the excavation was conducted 0.15 m apart from the roots and at a depth of 0.30 m. The excavated soybean roots were transported to wash off the soil using tap water mixed with mild detergent. The soil-free roots were drip-dried and suspended in front of a black background sheet and photographed using a digital camera (Nikon D3200, 6016 × 4000 pixels).

All photographs were uploaded to the DIRT platform to determine root traits, such as stem diameter (SDM, diameter at the beginning of the root top), root area (RA, the root area belonging to the root counted by pixels), root density (RD, the percentage of foreground to background pixels within root shape), median root system width (MEW, median root system width measured horizontally), maximum root system width (MAW, maximum root system width measured horizontally), basal root number (BRN, number of basal roots), and basal root angle (BRA, root angle between the soil horizontal reference and basal roots). After conducting photography, the roots were used to assess the root length (RL) by measuring the distance between the stem and the end of the taproot with a ruler, and root weight (RW, the total root weight), after performing drying at 70 °C for 48 h.

2.4. Soil macropore structure characteristics

The in situ dye-tracing method was applied to quantify the visual pore structure, as reported in several studies (Morris and Mooney, 2004; Sander and Gerke, 2007). The dye-tracing experiment was conducted for only five soybean varieties after harvesting them on 24th September 2017. As agricultural machinery has been increasingly used in arable lands since the 1970s, soil bulk density has consistently increased in the Mollisol region. According to the white paper of back soil (Chinese Academy of Sciences, 2021), soil bulk density increased by 7.59%, 34.18%, and 59.49% after reclamation for 20, 40, and 80 years, respectively. The varieties which were extensively promoted, before and after the period in the study region, were selected to represent the effects of soil compaction on plant breeding. The selected varieties included the following: Mancangjin, bred in 1941 (MCJ41); Hejiao, bred in 1963 (HJ63); Heinong, bred in 1975 (HN75); Suinong, bred in 1996 (SN96); and Hefeng, bred in 2003 (HF03) (see more information for the varieties in Supplementary Table 1). In total, ten treatments (two compaction levels × five soybean varieties) were selected, with three replications for each treatment.

The dye-tracing measurement protocol used in this study was adapted from that reported by Wang and Zhang (2017). However, brilliant blue was not in contrast with the black colour of Mollisol. Therefore, white TiO₂ powder (0.26–0.42 μm diameter) was added to improve the contrast of dye colours. This modification was performed according to the methods described by Liu and Lennartz (2015). First, a representative soybean plant was selected from each plot based on the stem diameter. Subsequently, the plant stem was centred in a PVC tube (0.20 m diameter and 0.35 m height), such that most roots were located underneath. Then, the PVC tube was pushed down to a depth of 0.15 m and filled with 1.6 L 10.0 g L⁻¹ TiO₂ suspension. To ensure identical infiltration conditions, plastic sheets were placed on the soil surface before utilisation of the suspension. After application of the suspension, the plastic sheets were rapidly removed to initiate the infiltration experiment and then placed on top of the tubes for 24 h to prevent evaporation. At the end of the infiltration, the tubes were further inserted into the soil with a rubber hammer to a depth of 0.30 m and were then excavated from the field.

The excavated soil columns were carefully packed and transported to the laboratory. Before acquiring images (with a digital camera, Nikon

D3200, 6016 × 4000 pixels), positions (east, west, north, south, and intervals of 10 mm depth) were marked on a wall of each PVC tube. Top-shooting photos were acquired in 10-mm intervals after the soil surface was prepared using a cutting machine (DongCheng, CWSM710-100). The exposed surfaces of the soil columns were levelled off using a knife and brushed carefully. Then, the photos were scaled using a 0.30 m × 0.30 m metal frame with reference to the marked positions. The obtained digital images were orthogonalised using the WGEO software (WASY GmbH) by transforming the four corners of the photographed metal frames from internal rasters to real coordinates. Thereafter, all images were loaded into the ImageJ software (National Institute of Mental Health, USA) and regions of interest (ROI, 0.18 m diameter and 0.30 m height) were selected. The undyed parts of the orthogonalised images were removed with the Photoshop 7.0 software. This was achieved by using the 'Colour range' function in 'Select' option, wherein the dyed parts were manually selected by conducting comparison with the original images. After segmentation, the images were transformed into binary format using the Image J software, and noise (isolated pixels) was removed to obtain a better image quality. As a result, the final pixel resolution was 0.3 × 0.3 mm for each slice.

The maximum dying depth (Inf_Max) was recorded when no dye traces could be detected, and the mean dye coverage (Cov_Mean) was calculated as the mean dye coverage within the ROI. The actual infiltration depths (H_a) (Eq. 1) were used to indicate the continuously dyed vertical distance according to the methods reported by Wang et al. (2006), and the horizontal spatial heterogeneity of macropores (Eq. 2–4) was calculated according to the methods reported by McGarigal and Marks (1995). A three-dimensional coordinate (X, Y, Z) was applied to the slices within one soil column, where the X (0.18 m), Y (0.18 m), and Z (0.3 m) axes were cropped along the diameter in the east-west and south-north directions, as well as the cylindrical axis of the soil column, respectively. The central position of the soil surface was set as the base point. To reduce the number of calculations, each horizontal plane (α) was divided into grids of 10 mm × 10 mm with a 10-mm cutting depth, which led to a volume resolution of 10 × 10 × 10 mm³. Then, the actual infiltration depth (H_a(x, y)) in the vertical direction of the column was calculated using the following equation:

$$H_a(x, y) = \sum_{i=1}^k \alpha(x, y, z) \Delta h_i \quad (1)$$

where, α(X, Y, Z) represents the coordinate of α grid, when the soil was dyed at the coordinates of (x, y, z), α(x, y, z) = 1; if not, α(x, y, z) = 0; Δh_i indicates the distance between the two adjacent grids (10 mm) in the Z-axis; and k represents the total number of slices. Note that H_a can be markedly smaller than the maximum dying depth, because of the heterogeneity of the soil structure.

Actual infiltration depths (H_a) in the grid images were then used to calculate horizontal spatial heterogeneity of macropores, using the FRAGSTATS program (McGarigal and Marks, 1995). It has been defined as the contagion index (CONTAG, Eq. 2), the patch cohesion index (COHESION, Eq. 3), and the aggregation index (AI, Eq. 4), which represent the fragmentation of macropore classes, horizontal macropore connectedness, and macropore aggregation in the horizontal layers (McGarigal and Marks, 1995). The contagion index can be described as:

$$\text{CONTAG} = \left[1 + \frac{\sum_{i=1}^m \sum_{k=1}^m \left[p_i \cdot \frac{g_{ik}}{\sum_{k=1}^m g_{ik}} \right] \left[\ln \left(p_i \cdot \frac{g_{ik}}{\sum_{k=1}^m g_{ik}} \right) \right]}{2 \ln(m)} \right] * 100\% \quad (2)$$

where p_i represents the proportion of the H_a grid occupied by class i; g_{ik} represents the number of adjacencies (joins) between pixels of classes i

and k , based on a double-count method, and m represents the number of classes present in the H_a grid, including the H_a grid border, if present. The cohesion index can be described as:

$$COHESION = \left[1 - \frac{\sum_{i=1}^m \sum_{j=1}^n p_{ij}^*}{\sum_{i=1}^m \sum_{j=1}^n p_{ij}^* \sqrt{a_{ij}^*}} \right] \left[1 - \frac{1}{\sqrt{Z}} \right]^{-1} * 100\% \quad (3)$$

where p_{ij}^* represents the perimeter of class patch ij in terms of number of cell surfaces, a_{ij}^* represents the area of class patch ij in terms of number of cells, and Z represents the total number of cells in the H_a grid. The aggregation index can be described as:

$$AI = \left[\sum_{i=1}^m \left(\frac{g_{ii}}{\max \rightarrow g_{ii}} \right) p_i \right] * 100\% \quad (4)$$

where g_{ii} represents the number of like adjacencies (joins) between pixels of class i based on the single-count method; $\max \rightarrow g_{ii}$ represents the maximum number of like adjacencies (joins) between pixels of class i based on the single-count method, and p_i represents the proportion of the H_a grid comprising of class i .

2.5. Statistical analysis

Two-way ANOVA was used to ascertain significant differences in the morphological traits of the roots and structural properties of the soil based on the growth of the five soybean varieties under the two compaction levels. Pearson correlation analysis was used to identify significant correlations between soil structural properties and root traits. Variance partitioning analysis was performed to further quantify the relative contributions of soil compaction (soil penetration resistance of the top 0.3 m layer) and root growth (median root system width) to changes in the soil structure. The variance partitioning analysis was performed with the use of “varpart” in the “vegan” packages, based on the redundancy analysis to express the explained variation. All statistical analyses were performed using the R software (The R Programming Language, <https://www.r-project.org/>). The significance threshold was set at $P \leq 0.05$.

3. Results

3.1. Physical properties of the soil

Mollisol demonstrated a relatively high organic matter content ($\sim 3\%$, Table 1) and a low bulk density and penetration resistance before soil compaction. The soil penetration resistance and bulk density between 15 and 35 cm of depth were significantly higher under compaction conditions than those observed under no compaction conditions (Fig. 1). The maximum change in penetration resistance was observed at a depth of 25 cm and increased from 0.89 MPa in uncompacted soil to 1.45 MPa in compacted soil, showing an increase of 63%. Correspondingly, at the depth of 25 cm, soil bulk density increased significantly from 1.27 to 1.49 g cm⁻³ ($P \leq 0.05$). Moreover, at the depth of 35 cm, the soil bulk density increased from 1.15 to 1.48 g cm⁻³.

3.2. Variations of root traits and their response to soil compaction

The root systems of all soybean varieties at the full flowering stage showed visible changes caused by soil compaction. Under conditions of soil compaction treatment, the roots were distributed more laterally and restricted in the vertical direction (Fig. 2). This was further verified by the significantly decreased root length in compacted soil, a phenomenon which was observed in all varieties (Supplementary Table 2); however, there was no significant difference among the varieties (Fig. 3). Similarly, in all varieties, there was a trend of decrease in root area (-14%) in compacted soil ($P \leq 0.10$). The analysis of variance showed significant changes in the root area, root density, and median root system width of five soybean varieties, irrespective of soil compaction treatments (Fig. 3 and Supplementary Table 2). There were no significant effects of soil compaction or its interaction with varieties on any measured root traits.

3.3. The response of soil macropore characteristics to soil compaction and soybean varieties

The TiO₂ staining patterns demonstrated that soil macropores were rich in the plough layer for all five varieties under no compaction conditions and were reduced to different extents under soil compaction conditions, except for the latest variety HF03 (Fig. 4). The vertical distribution of dye showed different responses to soil compaction among the soybean varieties (Fig. 5). Under no compaction conditions, the dye

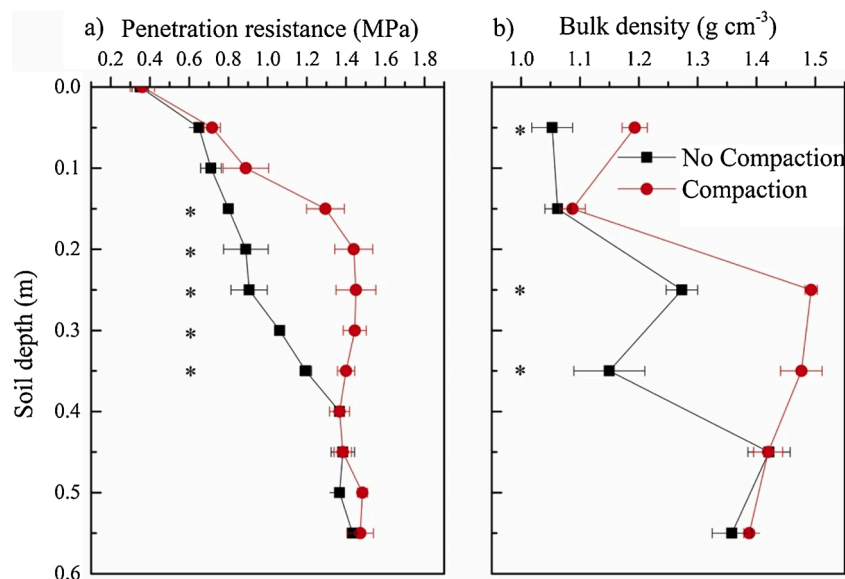


Fig. 1. Soil penetration resistance ($n = 30$) (a) and bulk density ($n = 4$) (b) within 0.55 m depth after six times of traffic wheeling at 60% of soil field capacity and surface rotary tillage. * indicates significant difference between the no compaction and compaction treatments at $P \leq 0.05$.

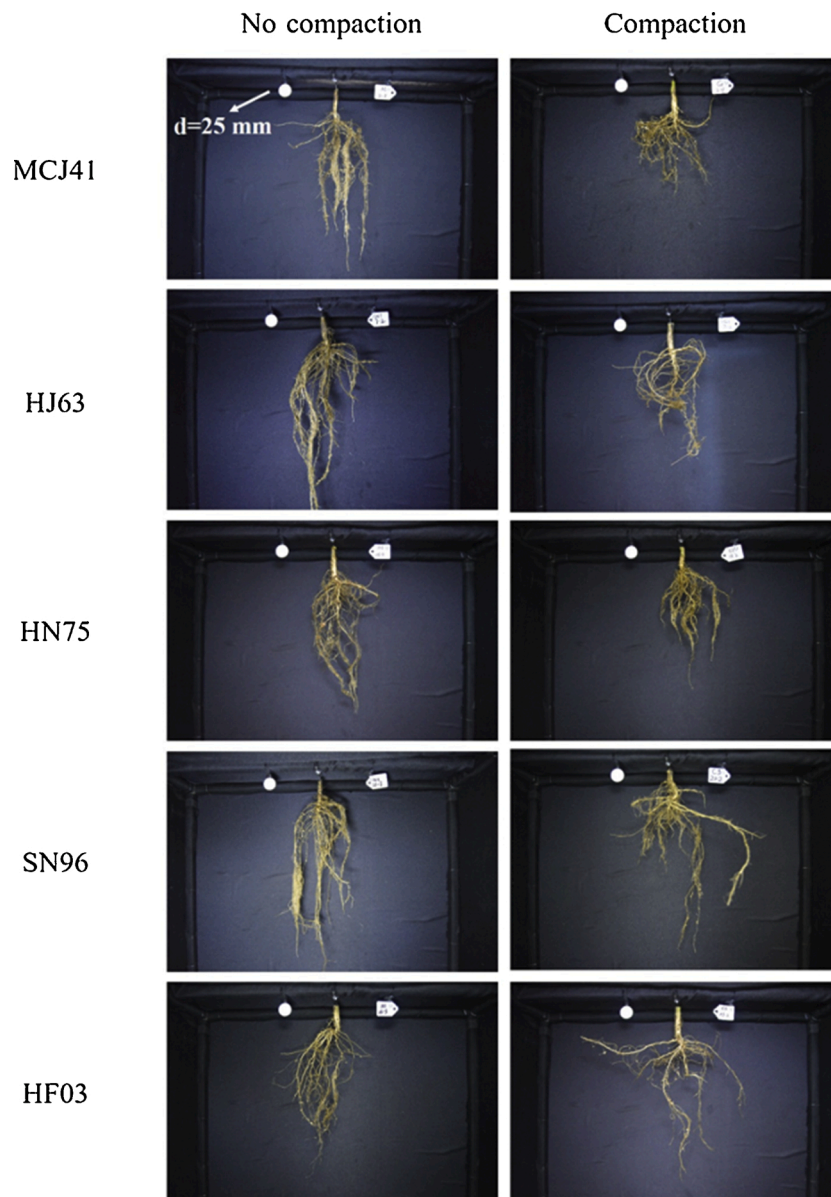


Fig. 2. Exemplified root images of five soybean varieties growth under the no compaction and compaction treatments. The white circles of 25 mm in diameter are used as size references.

coverage at the surface accounted for ~7% and decreased with increasing depth for HJ63 and SN96 varieties, while for the other three varieties, it accounted for 2%–4% and did not change with soil depth in the plough. Compared with no compaction conditions, soil compaction treatment led to decreases in the maximum dying depth and the dying coverage for all varieties, except for the most recent variety HF03. The HF03 had a higher dye coverage at the depth range of 0.44–10 cm and below 15 cm.

Vertical soil macroporosity and horizontal spatial heterogeneity were affected by soil compaction and its interaction with soybean varieties (Table 2). Soil compaction significantly decreased the maximum infiltration depth, mean dye coverage, and mean actual infiltration depth ($P \leq 0.05$). The effects of the interaction of soil compaction and soybean varieties were observable in the mean dye coverage and mean actual infiltration depth parameters ($P \leq 0.10$). Exemplified grid images (grid resolution of 10 mm $\times$ 10 mm) of the actual infiltration depth in the dyed soil columns are presented in Fig. 6. Based on the results obtained, soil compaction significantly increased the contagion and aggregation indices for all varieties at $P \leq 0.05$. Furthermore, the

interaction effects were observed in the aggregation index between soil compaction and soybean varieties, in which opposite trends were observed for the HF03 variety, although the difference did not reach statistical significance ($P \leq 0.05$).

3.4. Correlation between soil macropore structure and root traits

The vertical and horizontal structural properties were significantly ($P \leq 0.05$) correlated with each other, and they correlated well with the penetration resistance, median root system width, and maximum root system width (Fig. 7). The vertical structural parameters of soil, such as maximum dying depth, mean dye coverage, and mean dying depth, were negatively correlated with the soil penetration resistance and mean, as well as the maximum root width. The horizontal structural parameters of soil, such as the contagion index, patch cohesion index, and aggregation index, were positively correlated with the soil penetration resistance and median root system width.

Based on the correlation analysis, penetration resistance and median root system width were used to explain the variations in soil

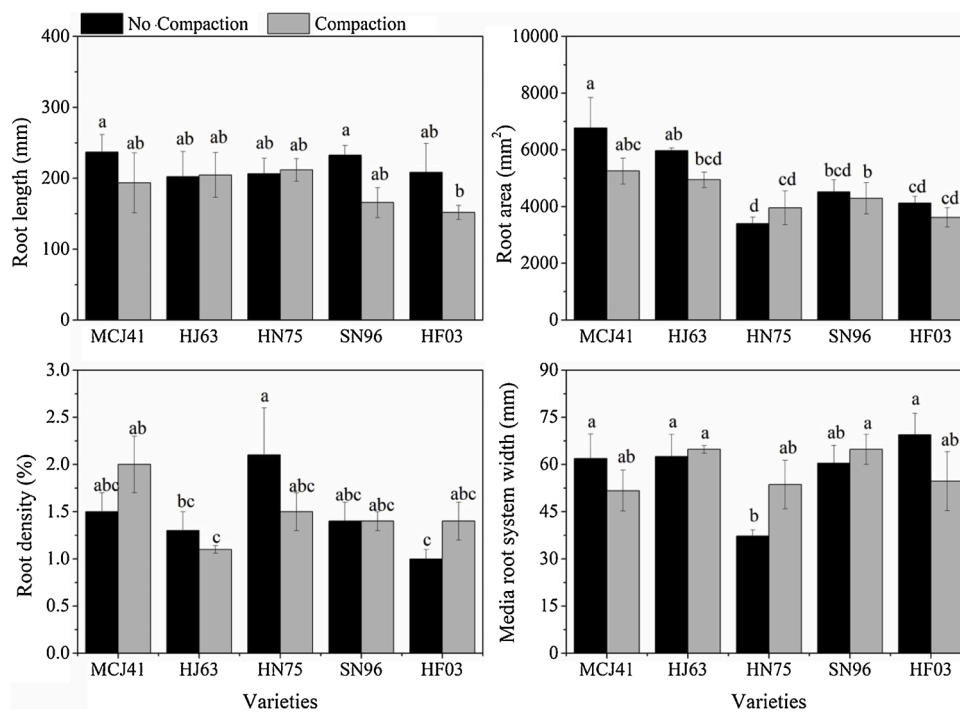


Fig. 3. Effects of soybean varieties and soil compaction on selected root traits. Bars indicate standard deviation of the means and different letters indicate significant difference at $P = 0.05$.

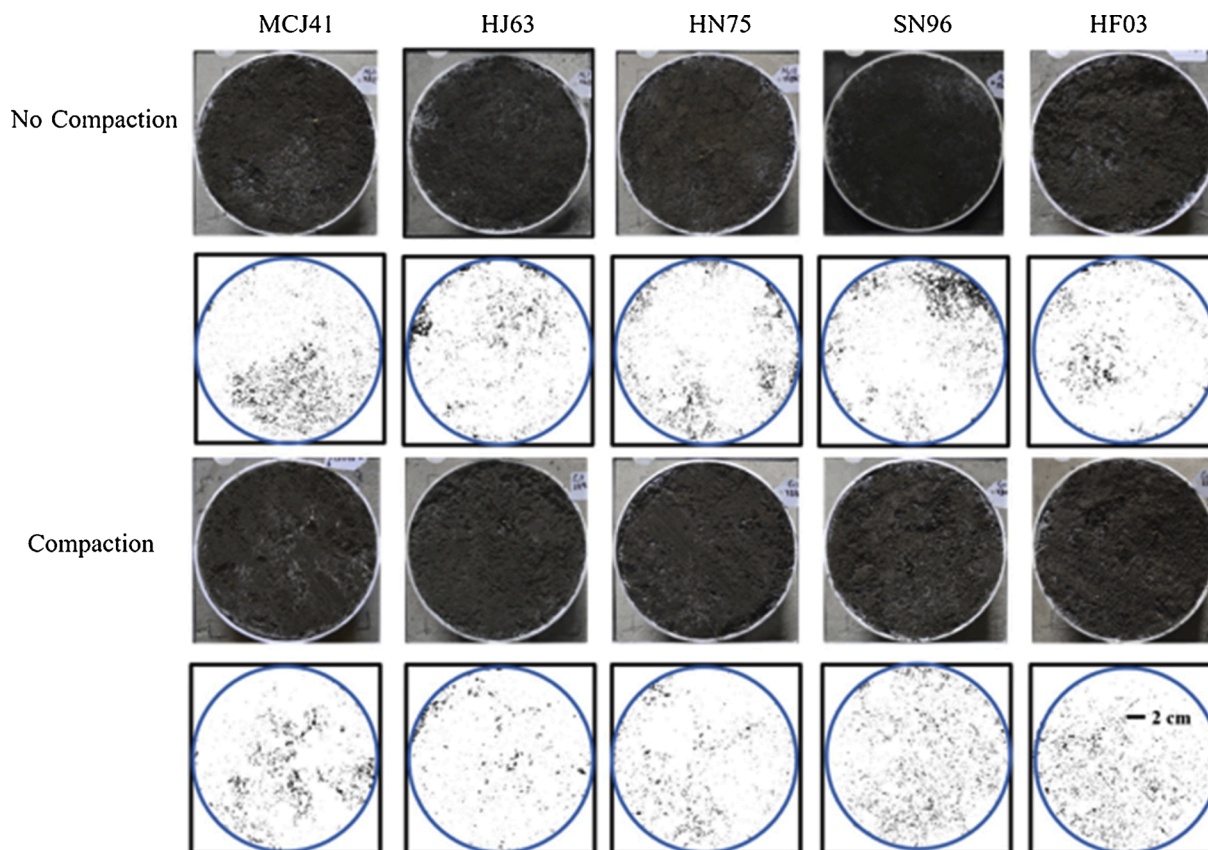


Fig. 4. Exemplified staining patterns of TiO_2 dye and corresponding binary images of soil slices at 0.06 m depth below soil surface with different soybean varieties under two soil conditions.

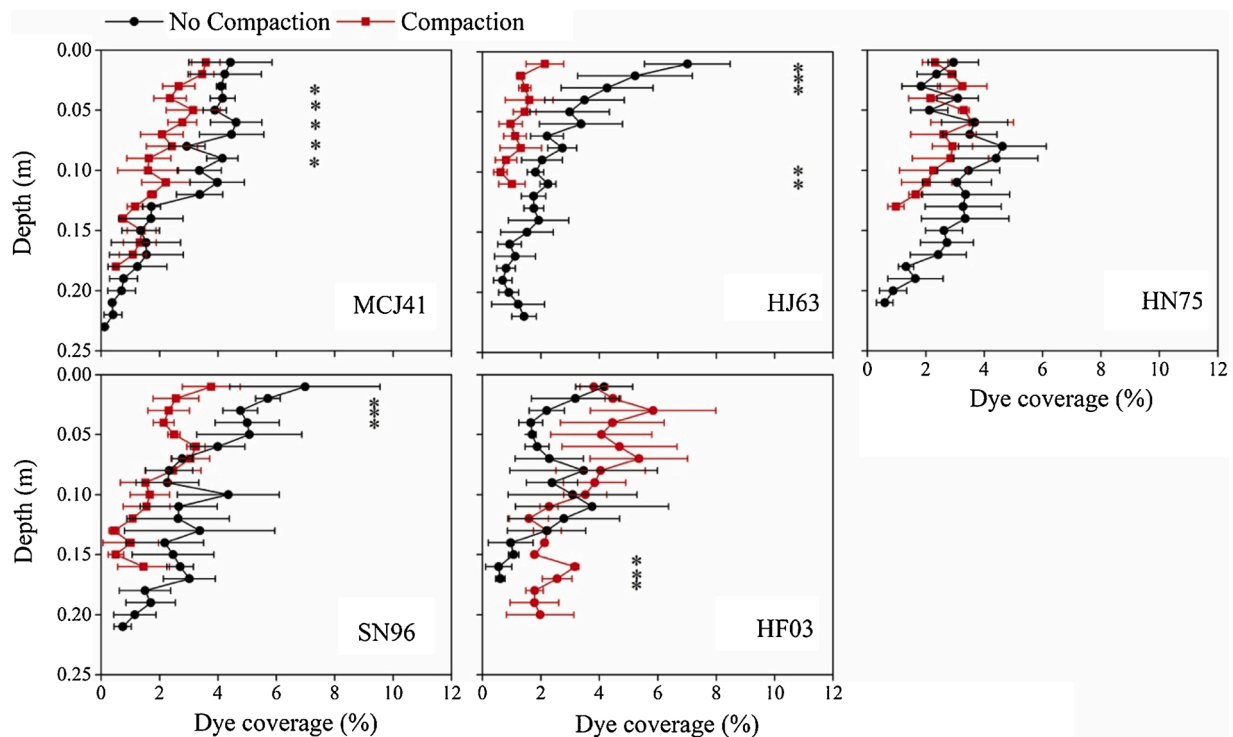


Fig. 5. Vertical dye coverage distribution in the soil columns planted with different varieties under soil compaction and no compaction conditions. Bars indicate standard deviation of the means of dye coverage ($n = 3$), and the asterisk indicates significant differences between the soil compaction and no compaction treatments ($P \leq 0.10$).

Table 2

Soil macroporosity and its horizontal spatial heterogeneity as affected by soybean varieties and soil compaction.

Factors		Macroporosity			Horizontal spatial heterogeneity		
		Inf_Max (mm)	Cov_Mean (%)	Ha_Mean (mm)	CONTAG (-)	COHESION (-)	AI (-)
Varieties (n = 6)							
	MCJ41	196.7 ± 22.2 a	1.5 ± 0.3 a	6.2 ± 1.3 a	35.8 ± 1.3 a	89.1 ± 2.1 a	46.4 ± 3.8 a
	HJ63	168.3 ± 26.6 a	1.1 ± 0.3 a	4.9 ± 1.8 a	40.2 ± 6.6 a	88.1 ± 4.4 a	56.1 ± 8.8 a
	HN75	166.7 ± 27.4 a	1.6 ± 0.3 a	6.9 ± 1.3 a	32.7 ± 2.8 a	84.6 ± 4.3 a	42.6 ± 3.7 a
	SN96	175.0 ± 23.1 a	1.6 ± 0.4 a	6.8 ± 1.7 a	34.2 ± 5.2 a	85.3 ± 3.3 a	45.9 ± 5.7 a
	HF03	173.3 ± 15.1 a	1.6 ± 0.4 a	6.7 ± 1.5 a	33.8 ± 4.5 a	83.7 ± 4.8 a	44.7 ± 5.2 a
Soil compaction (n = 15)							
	No compaction	205.3 ± 7.9 a	1.8 ± 0.2 a	7.8 ± 0.8 a	31.5 ± 2.0 a	83.4 ± 2.1 a	41.8 ± 2.2 a
	Compaction	146.7 ± 13.0 b	1.2 ± 0.2 b	4.8 ± 0.8 b	39.2 ± 2.8 b	88.9 ± 2.2 b	52.4 ± 3.9 b
Treatments (n = 3)							
MCJ41	No compaction	233.3 ± 8.8 a	2.0 ± 0.4 a	8.1 ± 1.5 a	33.5 ± 1.2 a	85.7 ± 2.3 a	42.0 ± 2.7 b
	Compaction	160.0 ± 25.2 a	1.1 ± 0.3 ab	4.2 ± 0.9 ab	38.1 ± 0.1 a	92.4 ± 1.1 a	50.8 ± 5.9 b
HJ63	No compaction	220.0 ± 11.5 a	1.7 ± 0.3 a	8.0 ± 1.6 a	28.8 ± 6.6 a	81.3 ± 5.7 a	40.4 ± 3.8 b
	Compaction	116.7 ± 12.0 a	0.5 ± 0.1 b	1.7 ± 0.6 b	51.5 ± 2.9 a	94.9 ± 1.0 a	71.7 ± 7.8 a
HN75	No compaction	210.0 ± 5.8 a	1.9 ± 0.4 a	8.4 ± 2.0 a	28.6 ± 3.6 a	80.0 ± 7.2 a	37.6 ± 4.1 b
	Compaction	123.3 ± 34.8 a	1.3 ± 0.2 ab	5.4 ± 1.1 ab	36.9 ± 1.1 a	89.2 ± 1.9 a	47.6 ± 3.8 b
SN96	No compaction	193.3 ± 21.9 a	2.1 ± 0.4 a	9.2 ± 1.9 a	30.0 ± 4.5 a	81.5 ± 3.4 a	39.0 ± 4.1 b
	Compaction	156.7 ± 37.6 a	1.0 ± 0.3 ab	4.4 ± 1.7 ab	38.4 ± 8.5 a	89.0 ± 4.5 a	52.9 ± 8.4 b
HF03	No compaction	170.0 ± 11.5 a	1.3 ± 0.4 ab	5.2 ± 1.5 ab	36.6 ± 5.0 a	88.4 ± 3.9 a	50.1 ± 6.9 b
	Compaction	176.7 ± 28.5 a	2.0 ± 0.6 a	8.2 ± 2.4 a	31.0 ± 7.2 a	79.0 ± 7.7 a	39.2 ± 6.0 b
P values							
Varieties		0.69	0.55	0.70	0.60	0.72	0.19
Compaction		<0.01**	0.02**	0.01**	0.02**	0.07*	0.01**
Varieties × Compaction		0.15	0.09*	0.08*	0.11	0.15	0.02**

Inf_Max, maximum infiltration depth; Cov_Mean, mean dye coverage; Ha_Mean, mean actual infiltration depth; CONTAG, contagion index; COHESION, patch cohesion index; AI, aggregation index. Different letters indicate significant differences at $P \leq 0.10$ (*) or $P \leq 0.05$ (**) among the varieties or between the soil compaction and no compaction treatments.

macropores, in both vertical and horizontal directions, in the variance partitioning analysis (Fig. 8). The penetration resistance and mean root system width explained 0.40 to 0.60 of the variations in the soil macropore structure. The soil compaction effects (0.278–0.384) were approximately three times more pronounced than the root growth effects (0.051–0.170). Additionally, root growth affected the vertical

distribution characteristics of soil macropores (0.116–0.170) more than horizontal heterogeneity (0.051–0.059).

4. Discussion

The studied Mollisol was low in bulk density and rich in organic

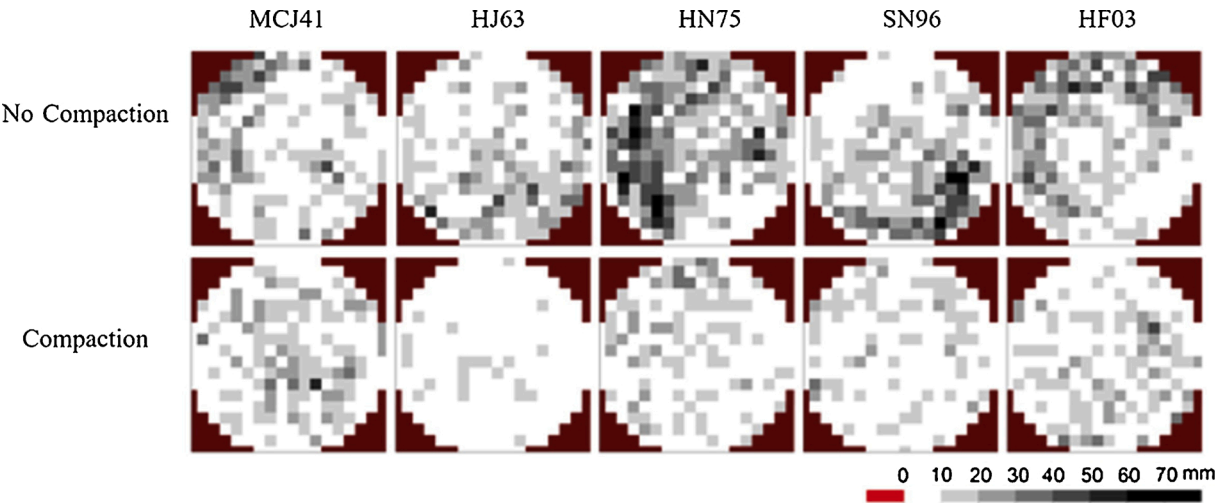


Fig. 6. Exemplified grid images (grid resolution of 10 × 10 mm) of actual infiltration depth (H_a) in the dyed soil columns planted with different soybean varieties under the compaction and no compaction conditions.

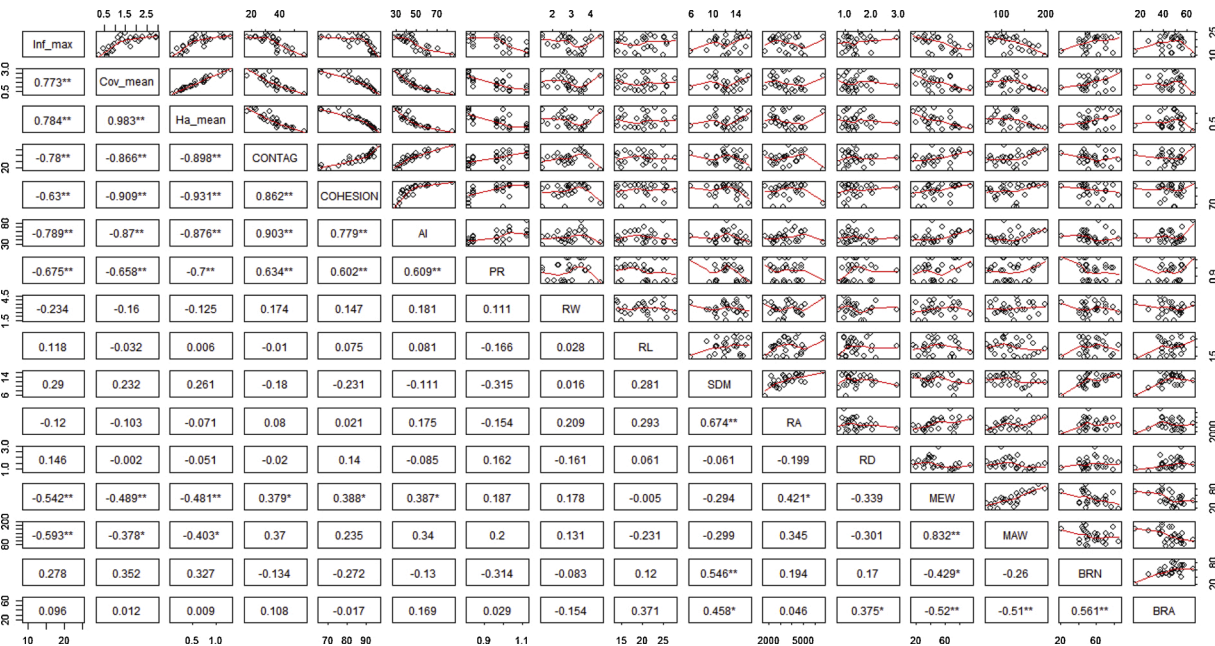


Fig. 7. Pearson correlations between soil structural properties and root traits of five soybean varieties ($n = 30$). Inf_Max, maximum dying depth; Cov_Mean, mean dye coverage; Ha_Mean, mean actual infiltration depth; CONTAG, contagion index; COHESION, patch cohesion index; AI, aggregation index; PR, penetration resistance; RW, root weight; RL, root length; SDM, stem diameter; RA, root area; RD, root density; MEW, median root system width; MAW, maximum root system width; BRN, basal root number; BRA, basal root angle. * and ** indicate significant correlations at 0.10 and 0.05 probability, respectively.

matter (Table 1), which led to significant soil compaction, particularly at the depth of 15–35 cm, after the six-pass traffic (Fig. 1). This was mainly attributed to the low soil strength because of the small soil bulk density (An et al., 2015; Keller et al., 2011). Mollisol with low soil strength could be easily compacted and this was also confirmed by other studies in this study region (Zhang et al., 2006; An et al., 2015). After the performance of rotary tillage, the topsoil was loosened; however, it did not completely alleviate the soil bulk density in the top 0–10-cm layer in the compaction treatments. This could be caused by the increased number of compacted clods after the rotary tillage in compacted soil. A similar phenomenon was observed in other regions with respect to different soil types (Arvidsson and Håkansson, 1996; Ren et al., 2019a).

Once the soil is subjected to compaction, root traits may exhibit changes. Our results showed that root length decreased by 17% in compacted soil, irrespective of the soybean variety (Supplementary

Table 2). However, within each variety, there was no significant difference in any of the selected root traits, between the plants grown under compacted and uncompacted soil conditions (Fig. 3). One probable reason could be the existence of a relatively humid weather during the growing season (Supplementary Fig. 1), which caused a high soil water content and led to a lower penetration resistance, even under compacted Mollisol conditions. Subsequently, this reduced differences in root traits between compacted and uncompacted soil conditions. Additionally, it is noteworthy that although Shovelomics is a fast and effective method, it could only screen the main root architecture and may omit the provision of information on fine roots. For instance, Clark (2008) has stated that root penetration ability is more closely related to root diameter, and Chimungu et al. (2015) have reported that root anatomical phenes are extremely sensitive indicators of root penetration resistance. Therefore, more detailed root evaluation methods should be conducted in

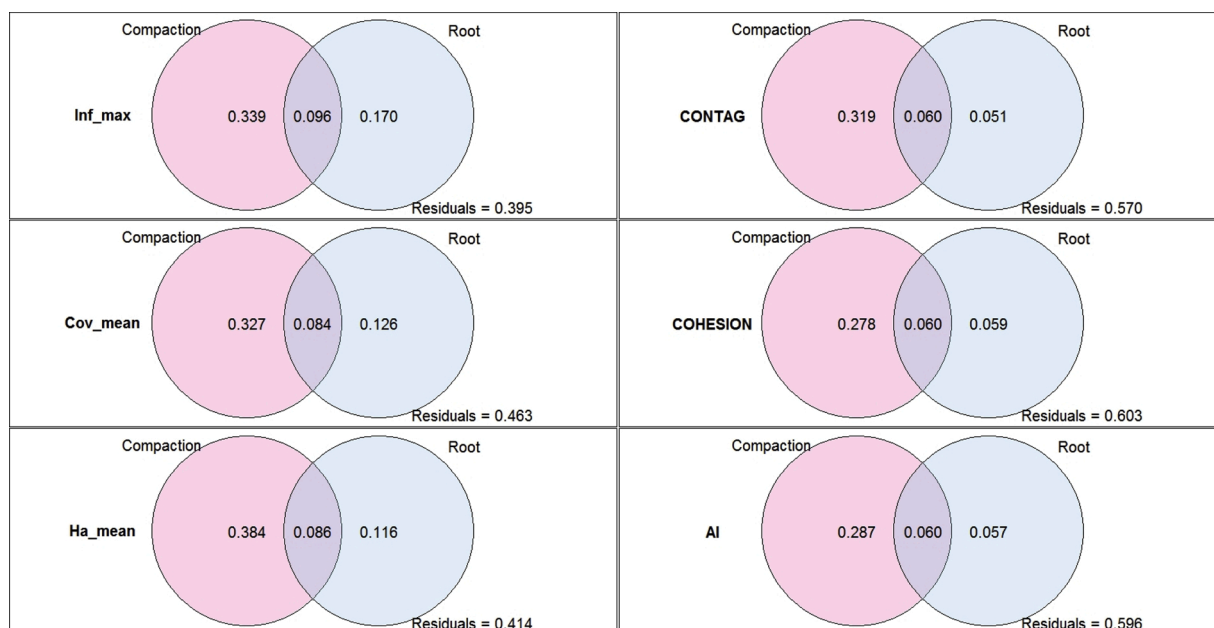


Fig. 8. Venn diagrams describing the relative effects of soil compaction (penetration resistance) and root growth (median root system width) on the vertical soil macroporosity and horizontal macropore heterogeneity in the soil columns. The circles in pink and sky-blue represent the portions of the variation accounted by soil compaction and root growth, respectively, with the overlapped part indicating the combined portion by these two factors ($n = 30$). The total effects of two factors on the variation of vertical macroporosity and horizontal macropore heterogeneity equals one subtracting residual (Inf_Max, maximum dying depth; Cov_Mean, mean dye coverage; Ha_Mean, mean actual infiltration depth; CONTAG, contagion index; COHESION, patch cohesion index; AI, aggregation index).

combination with the Shovelomics method to study the response of root traits to soil compaction in the future.

The dye tracing results showed distinctive root alleviation effects on the vertical distribution and spatial heterogeneity of the soil structure (Table 2 and Figs. 5 and 6). Under no compaction conditions, the dye showed changes in distribution related to increasing depth, which indicated various inherent root alleviation effects on soil macropore structure among the five selected varieties. Soil compaction reduced the maximum dying depths in the four old varieties, because of the loss of functional macropores due to soil compaction (Horn, 2003; Horn et al., 2000), while it increased the maximum dying depth under the latest variety HF03. This was related to an appreciable adaptability and plasticity of the roots of HF03 in response to soil compaction, which promoted deep root penetration to enable the creation of macropores. Soil compaction arranged soil particles and changed pore connectivity by utilising shearing force (Horn, 2003; Horn et al., 2000), which increased the spatial heterogeneity of the macroporosity of soil. This is true for the four old varieties, but not for the modern variety HF03 (Table 2). The decreased spatial heterogeneity in HF03 might be explained by the increased levels of root exudates, a phenomenon which assisted with the formation of stable soil aggregates (Hallett et al., 2010; Ikeda et al., 1997), as well as an increase in the number of pores and wider pore distribution connected to the lateral growth of the root (Helliwell et al., 2019). Based on the dye-tracing results, the latest varieties showed greater adaptability and plasticity in recovering the macropore structure. The different alleviation abilities among varieties might contribute to the soil compaction levels during breeding, because crops bred in compacted soils normally demonstrate higher resilience (Correa et al., 2019). However, further confirmation with more soybean varieties is necessary. Additionally, the dye tracing method cannot be considered as a direct approach for visualisation of soybean recovery ability and the measurement of soil porosity changes and related soil properties (e.g., soil air permeability and water conductivity) may lead to the obtaining of more comprehensive results.

Penetration resistance and median root system width were attributable for 40%–60% of the total variation in vertical and horizontal macropore structures (Fig. 8). The significant correlation of macropore

structure with penetration resistance and median (or maximum) root system widths suggests that a decrease in penetration resistance or median root system widths can effectively increase the depth and quantity of macroporosity in the vertical direction. Correspondingly, the horizontal macropore spatial heterogeneity could increase when less root grew in the horizontal direction (Fig. 7). Considering the strong multicollinearity between the median root system width and maximum root system width indices ($R^2 = 0.832$), the median root system width can be considered as the key root trait for the macropore structure. Interestingly, the effects of penetration resistance on macropore structure were approximately three times as those of median root system width, which implied that only one-third of the macropore structure degradation could be alleviated by planting soybean varieties with a beneficial median root system width. The median root system width also provided additional evidence to explain maximum dying depth, mean dye coverage, and mean actual infiltration depth (0.116–0.170) in a better manner than the contagion, patch cohesion, and aggregation indices (0.051–0.059). This indicated that the decreasing median root system width mainly alleviated the macropore structure in the vertical direction. However, we should consider that bio-alleviation is a slow process, and the absolute improvement may be limited within one growing season.

Root traits are crucial for crop growth, especially under conditions, such as the use of fertiliser, water, or subjection to mechanical stress. In the past, plant breeding programmes have focussed more on improving nutrient utilisation efficiency, rather than increasing nutrient acquisition efficiency, and the latter is more promising for modern crop breeding, as nutrient acquisition is more important in degraded soil (Lynch, 2007). Our results revealed that the median root system width was a key root parameter responsible for soil alleviation effects, particularly with respect to increasing vertical distribution and decreasing horizontal spatial heterogeneity of soil macroporosity. These results provide meaningful insights for soybean farmers, which may aid the recovery of degraded macropore structure of the soil caused by the use of heavy machinery. Moreover, these findings may help breeders in the development of strategies to improve the adaptability of the soybean varieties to soil compaction.

5. Conclusions

Root traits and soil macropores were characterised to determine the response of historical soybean varieties to soil compaction, as well as to determine the alleviation effects on compacted Mollisol. The selected five soybean varieties, bred from 1941 to 2003, showed substantial variations in root traits, although no evident breeding trends were detected in the Shovelomics-based root trait analysis. Under compacted soil conditions, macropore structures differed among soybean varieties, wherein the vertical infiltration depth and drying coverage were significantly inhibited in the varieties bred before agricultural machinery has been increasingly used in 1970s, such as the HJ63 and HN75, and the macropore horizontal spatial heterogeneity was increased in the variety of HJ63. In contrast, increased vertical development and improved spatial heterogeneity of soil macropores was observed in the HF03 variety under conditions of the soil compaction treatment. The characteristics of soil macropores were explained by soil compaction approximately three times larger than the root growth (median root system width), and root growth affected soil macropores more in the vertical direction than in the horizontal direction. Additionally, the median root system width could be considered as an adapted biological indicator in soybean breeding to alleviate compacted soils, and root growth could explain one-third of the compacted macropores, with the use of the dye-tracing method. Furthermore, it seems modern varieties presented a greater alleviation capacity, which indicated that plant breeders had unintentionally selected better varieties with improved root systems. However, further work with more soybean varieties within a longer breeding time span will be needed to corroborate this finding. The results obtained in this study will considerably benefit crop growth under prevailing climatic conditions, especially when encountering the issue of an increased frequency of droughts.

Declaration of Competing Interest

The authors report no declarations of interest.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:<https://doi.org/10.1016/j.still.2021.105166>.

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